

7. Seawater contains several dissolved salts.

- (a) The salinity of seawater is a measure of the total amount of dissolved salts and can be calculated using the following formula.

$$\text{Salinity (parts per thousand)} = \text{chloride ion concentration (mg l}^{-1}\text{)} \times 0.0018066$$

A sample of seawater had a salinity of 35 parts per thousand.

Calculate the chloride ion concentration, in g l^{-1} , of this sample.

2

- (b) Another sample of seawater was analysed to determine the mass of chloride ions present. The sample was titrated with a standard solution of silver nitrate solution.

(i) (A) State what is meant by a standard solution.

1

(B) Describe how a pipette should be prepared and used to accurately measure the sample of seawater.

2

- (ii) An indicator was added to the seawater in a conical flask.
State why an indicator is used.

1

[Turn over



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